475 SPECIAL PROBLEM SET #2

On pages 8 and 9 of the posted notes for this unit of study we used the grammar at the bottom of page 6 and the corresponding Shift-Reduce Matrix (SR) to determine if " a e a b # " was a valid string. This determination was made using our eleven-line parsing algorithm. It is important for you to take note of the fact that the parsing of that string was done after a considerable amount of analytical work had been done in order to derive SR. This is an example of what I have called the "95-5" rule. 95% of your work in SW development should be in analysis and 5% in writing the actual code. Most of the time those percentages are reversed. Do not let that happen to you and your co-workers, especially if you are in charge! Note: You should also attempt to develop the code with "pseudo-code" not an actual language such as JAVA or C++.

For this four point problem you are being asked to produce the trace as was done starting in the middle of page 8 and titled: CODE # ACTION STACK LOOKAHEAD. Your first line will be: 0 ____ 0 a a a e b a b a b #

You might also want to consider what the trace would look like if the "e" was omitted. Note: When we work with parsing problems such as this one, it is easy for a professor to always give problems where the input string is a valid. It is just as important to "see" what the parsing algorithm is doing when the input is not valid.

And finally, the syntactic definition for this "little" language is "little" for pedagogical reasons. The same kind of analysis and implementation was done for our very first "high-level" languages, probably FORTRAN and COBOL. You can imagine what the syntactic definition of those two languages must have looked like. Note: I was told that the original FORTRAN compiler took 7 years to write and debug, and that was with some 40 developers! TRUE? … I don't know, but it does not sound impossible given the amount of work we have done to handle our "little" language at the bottom of page 6!